



Gen AI and Agentic AI in Production

Week 3
Day 4

4 Week Journey to Prod-Grade

Week 4: Agentic AI in Production

Multi-Agent
Architecture

Capstone:
Agents on
AWS

Capstone:
Frontend

Observability
Monitoring
Security

Agentic AI
Platforms

Week 3: Azure, GCP, Data Engineering, Vectors, MCP

Cyber AI on
Azure with
MCP

Cyber AI on
GCP with
MCP

Amazon
SageMaker

Data Ingest
S3 Vectors

Data Pipes
with MCP

Week 2: AI Platform Engineering on AWS

AWS
Architecture

The Digital
Twin Mk 2

Amazon
Bedrock

Terraform

GitHub
Actions

Week 1: SaaS Generative AI in Production

First Gen AI
Deployment

Frontend &
Backend

Auth &
Subscription

Healthcare
SaaS App

Deploy to
AWS

Alex - the financial planner



Alex: The Agentic Learning Equities Explainer

What it is:

SaaS commercial application, financial planner

Set up:

Repo to clone: <https://github.com/ed-donner/alex>

Separate terraform directories

One environment, no Github Actions

Yesterday: SageMaker endpoint with embedding model

Bedrock vs SageMaker

Bedrock

SageMaker

Purpose

Use foundation models via API

Build, train, fine-tune, and deploy your own ML models

Models

Pre-trained models from Amazon and partners

Any model: bring your own, use open-source or prebuilt

Effort

Low - just call API, no infra to manage

Higher - manage infra, experiments, deployments

Deploy

Managed API endpoints

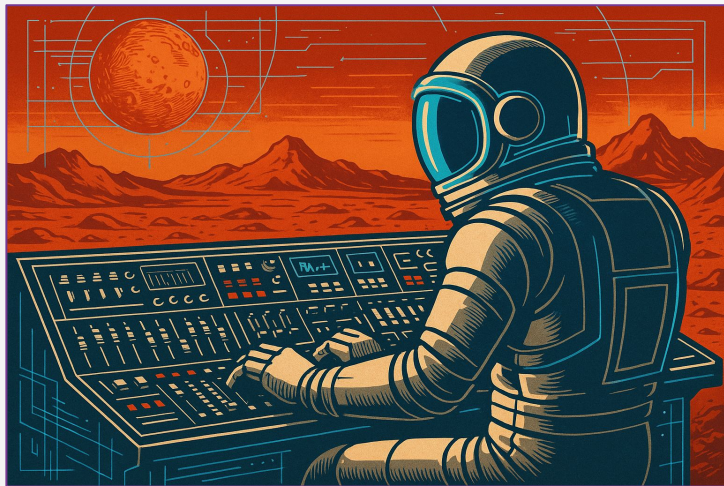
Scalable, customizable endpoints

Best for

Building with frontier models; inference at scale

End-to-end ML lifecycle, custom models

MLOps



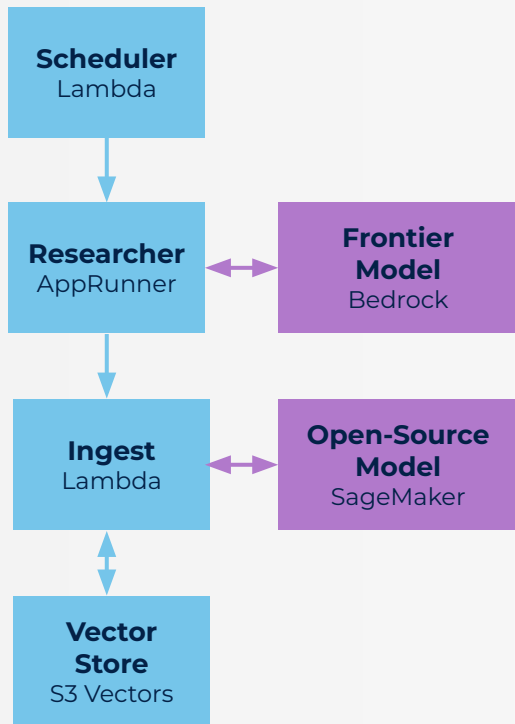
MLOps is Devops for ML: Manage the entire ML lifecycle

Track datasets, model versions, and training configurations so you can reproduce results and roll back if needed

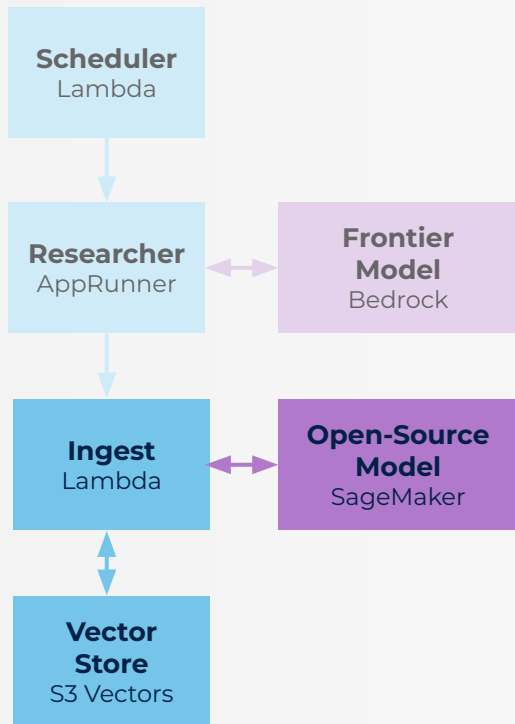
Monitoring & model drift – Deployed models can degrade over time as real-world data changes (“model drift”); MLOps involves monitoring predictions and retraining when accuracy drops.

SageMaker covers MLOps features: experiment tracking, model registry (for versioning), endpoints with monitoring, and automation tools for retraining when drift is detected.

Data Ingest Architecture



Data Ingest Architecture



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70%
Complete